Sheetal Jadhav

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RESEARCH INTERESTS

Aspiring computational neuroscientist with an Electronics Engineering background and hands-on experience across scales: from wet-lab developmental neurobiology and in vivo electrophysiology in songbirds to large-scale calcium imaging analysis and spiking network models of hippocampal learning. I build biologically grounded models of how plasticity shapes memory and behaviour, constrained by real neural data.

EDUCATION

National Institute of Technology, Karnataka (NIT-K), Surathkal, India — 2014 – 2018
Bachelor of Technology - Electronics and Communication Engineering — GPA: 6.4/10.00
Thesis Title: Implementation of a 10-bit resolution Delta-Sigma Analog-to-Digital Converter

Fr. Agnel School & Junior College, Vashi, Navi Mumbai — SSC 2011 · HSC 2013
Secondary School Certificate (SSC) & Higher Secondary Certificate (HSC) — GPA: 9.35/10 (SSC) & 8.4/10 (HSC)

SKILLS

- **Programming:** Python, MATLAB, C++, R
- **Wet Lab Techniques:** In vivo extracellular electrophysiology, Stereotaxic surgery & craniotomy, Custom electrode fabrication, Stereotaxic injections and lesioning (chemical and electrolytic), PCR, Cryo-sectioning, Histology, Immunohistochemistry (IHC)
- **Tools & Frameworks:** Brian2, NEURON, SpikeInterface, Bombcell, Pynapple, NeMoS, KlustaKwik

RESEARCH EXPERIENCE

Research Intern | **Campus Institute Data Science (CIDAS), University of Göttingen** — Remote
Supervisor: Dr. Tristan Manfred Stöber — Jan 2026 – Present

- Developed a proof-of-concept spiking network model of hippocampal latent-state learning: extended a Hebbian winner-take-all RNN with action-conditioned recurrent connectivity to approximate clone-structured cognitive graphs (CSCG), and designed a surprise-gated forward-prediction learning rule that resolves sensory aliasing by splitting ambiguous observations into context-specific clones
- Co-developed a DFG research proposal with my advisor for a biophysically grounded CSCG model in which multiple heterosynaptic plasticity mechanisms build latent-state representations in the hippocampus; formulated the central hypothesis, falsifiable predictions and a multi-scale work programme

Research Scholar | **Indian Institute of Science Education and Research (IISER)** — Pune, India
Supervisor: Dr. Raghav Rajan — Jan 2022 – Dec 2024

- Used in vivo electrophysiology in zebra finches to study pre-song neural dynamics and their role in vocal initiation and error correction; performed stereotaxic craniotomies, built custom electrodes and recorded from HVC and Area X (avian basal ganglia) in awake, singing birds
- Built a computational model of HVC to investigate what triggers introductory notes, the vocalizations that precede song

Research Consultant | **Janelia Research Campus** — Ashburn, VA, USA
Supervisor: Dr. Marius Pachitariu — Aug 2021 – Dec 2021

- Analysed calcium imaging data from 70,000 neurons to investigate how mouse visual cortex generalises across stimuli in stimulus-rich environments
- Conducted retinotopic analysis of 20,000 neurons in mouse visual cortex to understand sparse stimulus encoding

Visiting Research Fellow | **Homi Bhabha Centre for Science Education** — Mumbai, India
Supervisor: Prof. Sanjay Chandrasekharan — Jan 2020 – Jul 2021

- Investigated brain regions involved in face processing using seed-based functional connectivity analysis of the open-source Haxby fMRI dataset. Presented a poster at the 7th Annual Conference of Cognitive Science, IISc Bangalore (remote)

Junior Research Fellow | **Tata Institute of Fundamental Research (TIFR)** — Mumbai, India
Supervisor: Prof. Shubha Tole — May 2018 – Dec 2019

- Investigated the effects of overexpression and knockdown of Wnt signalling genes during development of the cortical hem (hippocampal organiser) in a mouse model; used bioinformatics and immunohistochemistry (IHC) to identify critical genes
- Identified differentially expressed genes through comparative analysis of transcriptomic datasets from the developing mouse brain

Summer Intern | **Indian Institute of Space Science and Technology (IIST)** Trivandrum, Kerala, India
Supervisor: Dr. Deepak Chopra Apr 2017 – Jul 2017

- Implemented an optimised watershed algorithm in MATLAB for segmentation of red blood cell images
- Predicted Alzheimer's disease from the open-source OASIS dataset using logistic regression, kNN, decision tree and support vector machine (SVM) classifiers

Summer Intern | **Indian Institute of Technology Bombay (IIT-B)** Mumbai, India
Supervisor: Prof. Y. S. Rao Apr 2016 – Jul 2016

- Calibrated polarimetric SAR systems using satellite imagery

AWARDS, FELLOWSHIPS & TRAINING

Impact Scholars 2026 Sep 2026
Selected as part of a team for a research proposal on how subjective-prior representations reorganise in ventrolateral OFC during adaptation to uncued context switches

Computational Approaches to Memory and Plasticity (CAMP) Jul 2022
3-week intensive summer school on computational models of memory and synaptic plasticity at National Centre for Biological Sciences (NCBS), Bangalore, India

Neuromatch Academy: Deep Learning course Jul 2023
3-week interactive online course
Neuromatch Academy: Computational Neuroscience course Jul 2021
3-week interactive online course

JGEEBILS Dec 2021
Joint Graduate Entrance Examination for Biology and Interdisciplinary Life Sciences:
qualified the national entrance examination for PhD programmes at leading Indian research institutes

Junior Research Scholarship, TIFR Oct 2018
Research fellowship funded by the Department of Atomic Energy (DAE), Govt. of India,
at the Tata Institute of Fundamental Research, Mumbai, India

JEE Main 2014 May 2014
Rank 3220 among ∼1.3 million candidates;
Secured admission to NIT Karnataka, one of India's top-ranked National Institutes of Technology.

Highest marks in Mathematics Jun 2013
Higher Secondary Certificate (HSC; final school-leaving exam), Maharashtra State Board

PEER REVIEW

Reviewer, NeurIPS 2026 Workshop – Foundation Models for the Brain and Body (BrainBodyFM)
brainbodyfm-workshop.github.io Sep 2026

TEACHING EXPERIENCE

- Lead Teaching Assistant; Computational Neuroscience course, Neuromatch Academy Jul 2026
- Teaching Assistant; Computational Neuroscience course, Neuromatch Academy Jul 2025
- Teaching Assistant; Animal Behaviour course, IISER Pune, for senior-year undergraduates (3 semesters) 2023 – 2024
- Course Grader; Developmental Neurobiology course, TIFR 2018 – 2019
- Volunteer Mathematics Teacher; training students from underrepresented backgrounds for Indian competitive exams at Savitribai Resource Centre, Navi Mumbai, India 2025 – Present

SELECTED COURSES

Graduate-level Courses

- Biostatistics
- Literature review and writing
- Mathematical and Computational Biology
- Neurobiology I & II
- Data Science and Practical Programming

Bachelor's Courses

- Pattern Recognition and Machine Learning
- Control Systems
- Digital Signal Processing
- Engineering Mathematics and Physics I & II
- Microprocessor and Embedded Systems

SCIENCE COMMUNICATION & OUTREACH

- Founder & Math Teacher, Prabuddha Vidyarthi STEM Network:
  a mentorship initiative for underprivileged students in India 2026 – Present
- Creator, Brainy Tales (Instagram, @onlybrainytales):
  neuroscience education content and the Neuroscience Scholars Collective Discord community 2025 – Present
- Author & Editor, First Principles (Substack): essays on science and politics 2026 – Present
- Pen pal, Letters to a Pre-Scientist: mentoring a middle-school student by letter 2026 – Present

REFERENCES

Prof. Aurnab Ghose
Professor, Department of Biology, Indian Institute of Science Education and Research (IISER), Pune, India
E-mail: aurnab@iiserpune.ac.in
Scholar Profiles: lab website

Dr. Raghav Rajan
Assistant Professor, Department of Biology, Indian Institute of Science Education and Research (IISER), Pune, India
E-mail: raghav@iiserpune.ac.in
Scholar Profiles: lab website